bigPLS: Fitting and Cross-Validating PLS based Cox Models to Censored Big Data

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Conclusion

The goal of `bigPLS` is to provide:

- easy to use functions to apply
- big data friendly functions
- sparse-group PLS-based Cox models
- fit and cross-validate regular, sparse, group or sparse group PLS regression models and their kernel counterparts to big data (Code1, Fig1)
- new robust measure based on Schmid score and R coefficient of determination
- improvement in selecting the number of components resulting in selecting better models (Fig2).

Availability

The R package bigPLS is available on Github: visit `https://github.com/fbertran/bigPLS/`.

Related works

The `bigPLS R package` implements or extends several algorithms that were published as two articles.

Deviance residuals-based sparse PLS and sparse kernel PLS regression for censored data, P. Bastien, F. Bertrand, N. Meyer, M. Maumy-Bertrand (2015), *Bioinformatics*, 31(3):397-404.

Fitting and Cross-Validating Cox Models to Censored Big Data With Missing Values Using Extensions of Partial Least Squares Regression Models, F. Bertrand and M. Maumy-Bertrand, *Frontiers in Big Data*, 4:684794, PMC8591675, (2021).

References

Bastien, P. (2008). Deviance residuals based PLS regression for censored data in high dimensional setting. Chemometrics and Intelligent Laboratory Systems 91, 78–86.

Cox, D. R. (1972). Regression models and life tables. Journal of the Royal Statistical Society. Series B: 34,187–220.

de Micheaux, P. L., Liquet, B., and Sutton, M. (2019). PLS for Big Data: A unified parallel algorithm for regularised group PLS. Statistics Surveys 13, 119 – 149.

Liquet, B., deMicheaux, P. L., Hejblum, B. P., and Thie´baut, R. (2015). Group and sparse group partial least square approaches applied in genomics context. Bioinformatics 32, 35–42.

Why PLS for censored big data?

Introduction

PLS has become an established tool in various experimental settings such as chemometric, networks, or systems biology. This modelling is primaliry used because it is often possible to explain the underlying system's extracted components and hence translate "hard" modelling information from the soft model: chemical components for NIR spectra, gene subnetwork for GRN or biological function for systems biology. As a consequence, choosing the right number of components is not only a major aim to avoid under or overfitting and ensure a relevant modeling or good predicting ability but also per se.

Relating personalized information from subjects such as omics data and subject survival or time to cancer recurrence is the focus of a vast literature from the last decade. The **discovery of markers from big data or high-dimensional data**, such as transcriptomic or SNP profiles, is a significant challenge in searching for more precise diagnoses. The most commonly used model for the analysis of survival data is the proportional hazard regression model suggested by Cox, 1972. Such a model helps the practitioner study in the presence of censoring the relationship between the time to event and a set of covariates. It has similar requirements as multivariate regression: more observations than variables, complete data, and not strongly correlated variables. In practice, when dealing with high-dimensional data, these constraints are crippling.

Fitting Cox models in a **big data context** -on a massive scale in terms of volume, intensity, and complexity exceeding the capacity of usual analytic tools-is often challenging. If some data are missing, it is even more difficult.

In this work, we deal with several PLS regression-based extensions of the Cox model that were first introduced in Bastien (2008) and Bastien et al. (2015) and extend them twice: **to group and sparse group models and to big data**. To archieve this goal, we use regular PLS models from Liquet et al. (2015) and de Micheaux et al. (2019). These extensions share features praised by practitioners, including regularization, interpretability of the components, missing data support, biplots of individuals and variables —and even parsimony for SPLS based models—, and allow to deal with highly correlated predictors or even rectangular datasets, which is especially relevant for high dimensional datasets.

In addition, we propose a **new robust measure** based on the Schmid score and the R coefficient of determination for least absolute deviation to assess Cox: the integrated R Schmid Score weighted (Fig2).

New available methods

Fitting Cox models on raw data

coxgpls
Fitting a Cox-Model on group PLSR components

coxsgpls
Fitting a Cox-Model on group sparse PLSR components

coxspls_sgpls
Fitting a Cox-Model on sparse PLSR components

coxbigsgpls
Fitting a sparse group PLS model with bigmemory support

coxDKbigsgpls
Fitting a Direct Kernel sparse group PLS model with bigmemory support

Using Deviance Residuals based models

coxgplsDR
Fitting a Cox-Model on group PLSR components

coxsgplsDR
Fitting a Cox-Model on group sparse PLSR components

coxspls_sgplsDR
Fitting a Cox-Model on sparse PLSR components

coxbigsgplsDR
Fitting a sparse group PLS model with bigmemory support

coxDKgplsDR
Fitting a Direct Kernel group PLS model

coxDKsgplsDR
Fitting a Direct Kernel group sparse PLS model

coxDKspls_sgplsDR
Fitting a Cox-Model on sparse PLSR components

coxDKbigsgplsDR
Fitting a Direct Kernel sparse group PLS model with bigmemory support

Cross Validating Models

cv.coxgpls
Cross-validating a Cox-Model fitted on group PLSR components

cv.coxgplsDR
Cross-validating a Cox-Model fitted on group PLSR components using (Deviance) Residuals

cv.coxsgpls
Cross-validating a Cox-Model fitted on sparse group PLSR components

cv.coxsgplsDR
Cross-validating a Cox-Model fitted on sparse group PLSR components using (Deviance) Residuals

cv.coxspls_sgpls
Cross-validating a Cox-Model fitted on sparse PLSR components

cv.coxspls_sgplsDR (Code1, Fig1)
Cross-validating a Cox-Model fitted on sparse PLSR components components using (Deviance) Residuals

Examples and Results

Code Snippet 1

Results of a cross-validation function

```
> (cv.coxspls_sgplsDR.res=cv.coxspls_sgplsDR(list(x=X_train_micro,time=Y_
    train_micro,status=C_train_micro),ind.block.x=c(3,10,15),nt=10))
$nt
[1] 10
$cv.error10
 [1] 0.5000000 0.6407988 0.6157086 0.6240256 0.5029069 0.5299969
 [7] 0.5125295 0.5812786 0.5835529 0.5958299 0.6275008
$cv.se10
 [1] 0.00000000 0.05351497 0.05687882 0.03957185 0.02526223
 [6] 0.04080999 0.04404444 0.04498782 0.02708915 0.02775493
[11] 0.02793903
$lambda.min10    #lambda.min: minimum value of lambda.
[1] 1
$lambda.1se10    #lambda.1se: largest value of λ such that error is within 1 standard
                 #error of the cross-validated errors for lambda.min.
[1] 0
```

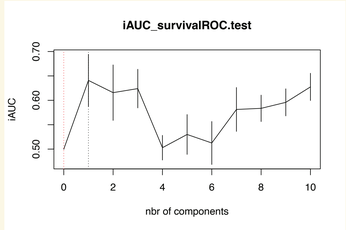

Figure 1: Displaying the AUCsurvROCtest criterion (mean and standard deviation) with respect to the number of components. The red dotted line id for the lambda.1se criterion and the black dotted line is for the lambda.min criterion.

Quality of Fit and Number of Components

New robust measure and improvement in selecting the number of components resulting in selecting better models

We used our recent results (see Figure 2) on using more relevant cross-validation criterion with the extensions to big data of our PLS based algorithms to set the cross-validation defaults in our packages. As can be seen for the AUCsurvROCtest criterion on Figure 2, the improvement of the performances due to switching to the recommended CV criteria is high enough to even have some (S)PLS based models, for instance SPLSDR, show some advantage over the other benchmark methods.

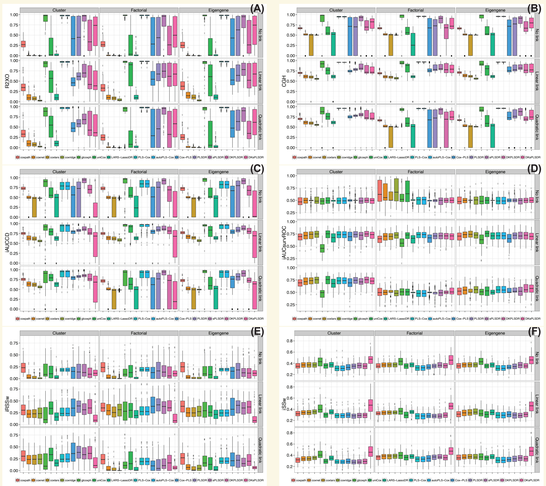

Figure 2: Panel by performance measure for models when fitted with AUCsurvROCtest cross validation panel (A): R2XO measure. panel (B): GHCI measure. panel (C): iAUCCD measure. panel (D): iAUCSurvROC measure, panel (E): the new iRSSw measure. panel (F): the new iSSw measure.